

Activity Sheet 3

Limits

Name: _____

Program/Section: _____

I. Complete the table and use the result to estimate the limit.

1. $\lim_{x \rightarrow 4} \frac{x-4}{x^2-3x-4} = \underline{\hspace{2cm}}$

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$						

2. $\lim_{x \rightarrow 2} \frac{x-2}{x^2-4} = \underline{\hspace{2cm}}$

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$						

3. $\lim_{x \rightarrow 0} \frac{\sqrt{x+6}-\sqrt{6}}{x} = \underline{\hspace{2cm}}$

x	-0.1	-0.01	-0.001	0.001	0.001	0.1
$f(x)$						

II. Find the limit of the following (if it exists).

4. $\lim_{x \rightarrow 2} x^3$

5. $\lim_{x \rightarrow 4} \sqrt[3]{x+4}$

6. $\lim_{x \rightarrow 2} \frac{\sqrt{x+2}}{x-4}$

7. $\lim_{x \rightarrow 7} \frac{3x}{\sqrt{x+2}}$

8. $\lim_{x \rightarrow 1} \cos \frac{\pi x}{3}$

9. $\lim_{x \rightarrow 0} \sec 2x$

10. $\lim_{x \rightarrow 3} \tan \frac{\pi x}{4}$

11. $\lim_{x \rightarrow \frac{\pi}{2}} \sin x$

12. $\lim_{x \rightarrow \pi} \tan x$

13. $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1}$

14. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

15. $\lim_{x \rightarrow 0} \frac{x}{x^2 - x}$

16. $\lim_{x \rightarrow 4} \frac{x - 4}{x^2 - 16}$

17. $\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x^2 - 9}$

18. $\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - 3}{x - 4}$

19. $\lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{x}$

20. $\lim_{x \rightarrow 0} \frac{\frac{1}{3+x} - \frac{1}{3}}{x}$